

Problem 1:

Let R be a ring, let $I \subset R$ be an ideal, and let $\mathfrak{p} \subset R$ be a prime ideal such that $I \subset \mathfrak{p}$. Prove that the collection of prime ideals

$$\{\mathfrak{q} \in \text{Spec}(R) \mid I \subset \mathfrak{q} \subset \mathfrak{p}\}$$

contains a minimal element.

Proof.

Let the set $\{\mathfrak{q} \in \text{Spec}(R) \mid I \subset \mathfrak{q} \subset \mathfrak{p}\}$ be ordered by super set relation. This is nonempty since $\mathfrak{p}$ is in it. Let Σ be a chain in that set.

Consider $\bigcap_{\mathfrak{q} \in \Sigma} \mathfrak{q} = A$. Let $ab \in A$.

Assume for contradiction that $a \notin A$ and $b \notin A$. Thus, there exists some $\mathfrak{q} \in \Sigma$ s.t. $a \notin \mathfrak{q}$ and $\mathfrak{q}' \in \Sigma$ s.t. $b \notin \mathfrak{q}'$.

Since there are a chain either $\mathfrak{q} \subset \mathfrak{q}'$ or $\mathfrak{q}' \subset \mathfrak{q}$. WLOG let $\mathfrak{q} \subset \mathfrak{q}'$.

This implies that $ab \notin \mathfrak{q}$ since $a \notin \mathfrak{q}$ and $b \notin \mathfrak{q}' \implies b \notin \mathfrak{q}$. This is a contradiction that $ab \in A$.

Therefore, this implies that either $a \in A$ or $b \in A$. Thus, A is prime.

This is an upper bounder under the super set relation. Thus, by Zorn's lemma we have that there exists a maximal element in $\{\mathfrak{q} \in \text{Spec}(R) \mid I \subset \mathfrak{q} \subset \mathfrak{p}\}$ which under normal inclusion relation is a minimal element. ■

Problem 2:

Let $f : R \rightarrow S$ be a ring homomorphism.

(a)

If $\mathfrak{p} \subset R$ is a minimal prime ideal which is contained in the image of the map $\text{Spec}(f) : \text{Spec}(S) \rightarrow \text{Spec}(R)$, prove that $\mathfrak{p}$ is the image of a minimal prime ideal in S .

Proof.

Assume $\mathfrak{p} \in \text{Im}(\text{Spec}(f))$. This implies that $f^{-1}(\mathfrak{q}) = \mathfrak{p}$ for $\mathfrak{q} \in \text{Spec}(S)$. By 1, this implies that there exists a minimal prime n s.t. $n \subset \mathfrak{q}$.

Thus, $f^{-1}(n) \subset f^{-1}(\mathfrak{q}) = \mathfrak{p}$, but since $\mathfrak{p}$ is minimal this implies that $f^{-1}(n) = \mathfrak{p}$. ■

(b)

If f is injective, prove that every minimal prime ideal of R is contained in the image of the map $\text{Spec}(f) : \text{Spec}(S) \rightarrow \text{Spec}(R)$, and give an example showing that this can fail if f is not injective.

Proof.

Assume that f is injective this implies that R can be considered a subring in S , so consider f just the inclusion map.

Let $\mathfrak{p}$ be a minimal prime in $\text{Spec}(R)$. Assume for contradiction that $\text{Spec}(f)^{-1}(\mathfrak{p}) = \emptyset$. In homework 4, we showed that $\text{Spec}(S_{\mathfrak{p}}/\mathfrak{p}S_{\mathfrak{p}}) \cong \text{Spec}(f)^{-1}(\mathfrak{p})$.

This implies that $S_{\mathfrak{p}}/\mathfrak{p}S_{\mathfrak{p}} = (0)$. Thus, $S_{\mathfrak{p}} = \mathfrak{p}S_{\mathfrak{p}}$. This implies that $a/s = 1/1$ for $a \in \mathfrak{p}$ and $s \in R \cap (R \setminus \mathfrak{p})$. This implies that there exists a $r \in R \cap (R \setminus \mathfrak{p})$ s.t. $r(a - s) = 0$ thus $ra = rs$. This implies that $rs \in \mathfrak{p}$ but $r \notin \mathfrak{p}$ and $s \notin \mathfrak{p}$ this is a contradiction, thus $\text{Spec}(f)^{-1}(\mathfrak{p}) \neq \emptyset$. Therefore, $\mathfrak{p}$ is in the image of $\text{Spec}(f)$, so by (a) we have that $\mathfrak{p}$ is in the image of a minimal prime ideal in S .

Example:

Consider $R = k[x, y]/(xy)$.

Consider $f : R \rightarrow R/(x) \cong k[y]$. This is surjective and not injective.

(y) does not lie in the image since $\text{Im}(\text{Spec}(f)) \cong V((x))$, and (y) is not contained in $V((x))$. ■

Problem 3:

For each of the following statements, either provide an example or prove that none exists:

(a)

A $\mathbb{Z}$ -module M whose support is not closed.

Proof.

Consider $M = \bigoplus_{p \text{ prime}} \mathbb{Z}/p\mathbb{Z}$.

$(\mathbb{Z}/p\mathbb{Z})_{(q)} = 0$ if $p \neq q$ since p is a unit in $\mathbb{Z}_{(q)}$, and if $p = q$ then $(\mathbb{Z}/p\mathbb{Z})_{(q)} \cong \mathbb{Z}/p\mathbb{Z}$ since everything is already invertible.

This implies that $\text{Supp}(M) = \{(p) \in \text{Spec}(\mathbb{Z}) \mid p \text{ is prime}\}$.

This set is not closed since every closed subset of $\text{Spec}(\mathbb{Z})$ is $V(n)$ for $n \in \mathbb{Z}$.

If $n \neq 0$, then $V(n)$ contains all the prime ideals (p) s.t. $p \mid n$, so $V(n)$ is finite.

If $n = 0$, then $V(0) = \text{Spec}(\mathbb{Z})$, but $(0) \notin \text{Supp}(M)$, so we have that $\text{Supp}(M)$ is not closed. ■

(b)

A ring R and a R -module M such that the support $\text{Supp}(M) \subset \text{Spec}(R)$ is not stable under specialization.

Proof.

False, let R be a ring and M a R -module. Let $\mathfrak{p} \in \text{Supp}(M)$ and $\mathfrak{q} \in V(\mathfrak{p}) \iff \mathfrak{p} \subset \mathfrak{q}$.

This implies that $M_{\mathfrak{p}} \neq 0$.

Since $R \setminus \mathfrak{q} \subset R \setminus \mathfrak{p} \implies M_{\mathfrak{p}} \cong S^{-1}M_{\mathfrak{q}}$ where $S = R \setminus \mathfrak{p}$.

Therefore, if $M_{\mathfrak{q}} = 0$ then this implies that $S^{-1}M_{\mathfrak{q}} = 0$, so $M_{\mathfrak{p}} = 0$, so by contrapositive we have that since $M_{\mathfrak{p}} \neq 0$ we know that $M_{\mathfrak{q}} \neq 0$. Therefore, $\mathfrak{q} \in \text{Supp}(M)$.

Thus, the $\text{Supp}(M)$ is stable under specialization. ■

Problem 4:

Let M be a module over a ring R , and let $S \subset R$ be a multiplicative system. Let $\varphi : \operatorname{Spec}(S^{-1}R) \rightarrow \operatorname{Spec}(R)$ be the injection induced by the canonical homomorphism $R \rightarrow S^{-1}R$. Prove that

$$\operatorname{Supp}_{S^{-1}R}(S^{-1}M) = \{S^{-1}\mathfrak{p} \mid \mathfrak{p} \in \operatorname{Supp}_R(M), \mathfrak{p} \cap S = \emptyset\} = \varphi^{-1}(\operatorname{Supp}_R(M)),$$

Proof.

$$S^{-1}\mathfrak{p} \in \varphi^{-1}(\operatorname{Supp}_R(M)) \iff \exists \mathfrak{p} \in \operatorname{Supp}_R(M) \text{ s.t. } \varphi^{-1}(\mathfrak{p}) = S^{-1}\mathfrak{p}$$

$\iff \mathfrak{p} \cap S = \emptyset$ and $\mathfrak{p} \in \operatorname{Supp}_R(M)$. This iff comes from the bijection between primes in $\operatorname{Spec}(S^{-1}R)$ and primes in $\operatorname{Spec}(R)$ disjoint from S .

$$\text{Thus, } \{S^{-1}\mathfrak{p} \mid \mathfrak{p} \in \operatorname{Supp}_R(M), \mathfrak{p} \cap S = \emptyset\} = \varphi^{-1}(\operatorname{Supp}_R(M))$$

For any prime $\mathfrak{p} \in \operatorname{Spec}(R)$ and $\mathfrak{p} \cap S = \emptyset$. This implies that S is already invertible in $M_{\mathfrak{p}}$. Thus, $(S^{-1}M)_{S^{-1}\mathfrak{p}} \cong M_{\mathfrak{p}}$. If $\mathfrak{p} \cap S \neq \emptyset$ then $S^{-1}\mathfrak{p} = (0)$.

$$\text{Therefore, } S^{-1}\mathfrak{p} \in \operatorname{Supp}_{S^{-1}R}(S^{-1}M) \iff \mathfrak{p} \in \operatorname{Supp}_R(M) \text{ and } \mathfrak{p} \cap S = \emptyset. \quad \blacksquare$$

Problem 5:

Let M be a module over a ring R , and let $S \subset R$ be a multiplicative system.

(a)

Prove that there is an inclusion

$$\{S^{-1}\mathfrak{p} \mid \mathfrak{p} \in \text{Ass}_R(M), \mathfrak{p} \cap S = \emptyset\} \subset \text{Ass}_{S^{-1}R}(S^{-1}M).$$

Proof.

Let $S^{-1}\mathfrak{p} \in \{S^{-1}\mathfrak{p} \mid \mathfrak{p} \in \text{Ass}_R(M), \mathfrak{p} \cap S = \emptyset\}$. This implies that $\mathfrak{p} = \text{Ann}_R(m)$ for $m \in M$. Consider $\text{Ann}_{S^{-1}R}(m/1)$. Let $a/s \in S^{-1}\mathfrak{p}$ then $a/s \cdot m/1 = am/s = 0$ since $am = 0$. Thus, $a/s \in \text{Ann}_{S^{-1}R}(m/1)$.

Let $a/s \in \text{Ann}_{S^{-1}R}(m/1)$. This implies that $a/s \cdot m/1 = am/s = 0$. Thus, this implies there exists a $t \in R \setminus \mathfrak{p}$ s.t. $tam = 0$. This implies that $ta \in \text{Ann}_R(m) = \mathfrak{p} \implies a \in \mathfrak{p}$. Thus, $a/s \in S^{-1}\mathfrak{p}$.

Therefore, $\text{Ann}_{S^{-1}R}(m/1) = S^{-1}\mathfrak{p}$. Thus, $S^{-1}\mathfrak{p} \in \text{Ass}_{S^{-1}R}(S^{-1}M)$. ■

(b)

If R is noetherian, prove that the above inclusion is an equality.

Proof.

Assume R is noetherian. Let $S^{-1}\mathfrak{p} \in \text{Ass}_{S^{-1}R}(S^{-1}M)$. This implies that $S^{-1}\mathfrak{p} = \text{Ann}_{S^{-1}R}(m/s)$ for $m/s \in S^{-1}R$. Since R is noetherian this implies that $\mathfrak{p} = (a_1, \dots, a_n)$. Thus, $S^{-1}\mathfrak{p} = (a_1/1, \dots, a_n/1)$.

This implies that $a_i/1 \cdot m/s = 0$ for all i .

Thus, there exists a $c_i \in R \setminus \mathfrak{p}$ s.t. $c_i a_i m = 0$.

Consider $\text{Ann}_R(c_1 c_2 \dots c_n m)$

Since $a_i \cdot c_1 \dots c_n m = 0 \implies (a_1, \dots, a_n) = \mathfrak{p} \subset \text{Ann}_R(c_1 \dots c_n m)$

Let $x \in \text{Ann}_R(c_1 \dots c_n m)$.

Thus, $xc_1 \dots c_n m = 0 \implies x/1 \cdot m/s = xm/s = \frac{xc_1 \dots c_n m}{c_1 \dots c_n s} = 0 \implies x/1 \in S^{-1}\mathfrak{p} \implies x \in \mathfrak{p}$.

Thus, $\mathfrak{p} = \text{Ann}_R(c_1 \dots c_n m)$, so $\mathfrak{p} \in \text{Ass}_R(M)$, and since there is a bijection of primes in $S^{-1}R$ and primes in R that don't meet $\mathfrak{p} \cap S = \emptyset$. We know that $\mathfrak{p} \cap S = \emptyset$.

Therefore, $S^{-1}\mathfrak{p} \in \{S^{-1}\mathfrak{p} \mid \mathfrak{p} \in \text{Ass}_R(M), \mathfrak{p} \cap S = \emptyset\}$. ■

Problem 6:

An embedded prime of a ring R is an associated prime of R (regarded as an R -module) which is not minimal among the associated primes of R .

(a)

Prove that if R is reduced, then R does not have any embedded primes.

Proof.

Assume R is reduced. Let $\mathfrak{p} \in \text{Ass}_R(R)$. This implies that $\mathfrak{p} = \text{Ann}(r)$ for $r \in R$.

Thus, $ra = 0$ for all $a \in \mathfrak{p}$.

Assume for contradiction there exists a prime ideal $\mathfrak{q}$ s.t. $\mathfrak{q} \subsetneq \mathfrak{p}$.

This implies that there exists a $a \in \mathfrak{p} \setminus \mathfrak{q}$ s.t. $ra = 0 \implies r \in \mathfrak{q}$.

Therefore, $r^2 = 0$ since $r \in \mathfrak{q} \subsetneq \mathfrak{p}$. Thus, $r \in \sqrt{(0)} = (0) \implies r = 0$. This is a contradiction since $\text{Ann}(0) = \text{Spec}(R)$.

Thus, $\mathfrak{p}$ is a minimal prime. ■

(b)

Give an example of a nonreduced ring which does not have any embedded primes.

Proof.

Consider $k[x, y]/(xy^2)$. This is not reduced since $(xy)^2 = 0$, and $xy \neq 0$.

Shown in class since $k[x, y]$ is a UFD and being quotiented by a principal ideal this implies the associated primes are primes that divide xy^2 .

Thus, $\text{Ass}(k[x, y]/(xy^2)) = \{(x), (y)\}$, and these are both minimal primes. ■

(c)

Prove that if $\mathfrak{p}$ is an embedded prime of R , then the localization $R_{\mathfrak{p}}$ is nonreduced.

Proof.

Assume $\mathfrak{p}$ is an embedded prime of R . This implies that $\mathfrak{p} = \text{Ann}(r)$ for $r \in R$. Therefore, in $R_{\mathfrak{p}}$ we have that $\mathfrak{p}R_{\mathfrak{p}} = \text{Ann}(r/1)$.

Assume for contradiction that $R_{\mathfrak{p}}$ is reduced.

Let $\mathfrak{q} \subsetneq \mathfrak{p}R_{\mathfrak{p}}$ be a prime ideal. This exists since if it doesn't, it would imply that $\mathfrak{p}$ is a minimal prime.

Thus, there exists a $a/s \in \mathfrak{p}R_{\mathfrak{p}} \setminus \mathfrak{q}$ s.t. $r/1a/s = 0 \implies r/1 \in \mathfrak{q}$.

Therefore, $r^2/1 = 0 \implies r/1 = 0$ since $R_{\mathfrak{p}}$ is reduced. This implies there is a $t \in R \setminus \mathfrak{p}$

s.t. $tr \in R \setminus \mathfrak{p}$, but this is a contradiction since it was shown that $r/1 \in \mathfrak{p}R_{\mathfrak{p}} \implies r \in \mathfrak{p}$.

Thus, $tr \in \mathfrak{p}$.

Therefore, $R_{\mathfrak{p}}$ is not reduced. ■

Problem 7:

Let M be a module over a noetherian ring R . Let $\text{Ass}^{\max}(M)$ denote the set of maximal elements in $\text{Ass}(M)$; note that this set is nonempty since R is noetherian. Prove that the homomorphism

$$\varphi : M \rightarrow \prod_{p \in \text{Ass}^{\max}(M)} M_p, \quad m \mapsto (f_p(m))_{p \in \text{Ass}^{\max}(M)}$$

is injective, where $f_p : M \rightarrow M_p$ is the localization homomorphism.

Proof.

Let $m \in \ker(\varphi)$. Thus, $\varphi(m) = 0$. Thus, $m/1 = 0 \in M_p$ for all $p \in \text{Ass}^{\max}(M)$.

This implies there exists a $c_p \notin p$ s.t. $c_p m = 0$. Let $J = \text{Ann}_R(m)$.

If $m = 0$ then done.

If $m \neq 0$ then $J \neq R$, so it is contained in some maximal ideal.

Consider the R -linear map $\varphi : R \rightarrow Rm, r \mapsto rm$.

This is surjective since every element of Rm looks like rm , and its kernel is J since if $r \in \ker(\varphi) \iff rm = 0 \iff r \in J$.

Thus, by first isomorphism theorem we get that $R/J \cong Rm$. Thus, since R is noetherian we know that $\text{Ass}(Rm)$ is nonempty since $Rm \neq 0$ since $m \neq 0$.

Let $p \in \text{Ass}(Rm)$. Since Rm is a submodule shown in class we know that $p \in \text{Ass}(M)$. Since $\text{Ass}(M)$ is nonempty and R is noetherian we know that there exists a maximal element q s.t. $p \subset q$.

Since $p \in \text{Ass}(Rm) \implies \exists a \in Rm$ s.t. $\text{Ann}(a) = p$. If $a/1 = 0$ in $(Rm)_q$, then we have some $s \in R \setminus q$ s.t. $sa = 0$. This implies that $s \in \text{Ann}(y) = p \subset q$ which is a contradiction. Therefore, $a/1 \neq 0$, so $(Rm)_q \subset M_q$ are not zero.

However, since $m/1 = 0 \in M_q$ this implies that for any $a = mx \in Rm$ that $a/1 = (r/1)(m/1) = 0$. This implies that $(Rm)_q$ is zero which is a contradiction.

Therefore, $m = 0$, so the kernel is trivial, so φ is injective. ■